# PHAGENT_AMR Workflow Documentation

This document provides visual documentation of the assembly-based AMR profiling pipeline.

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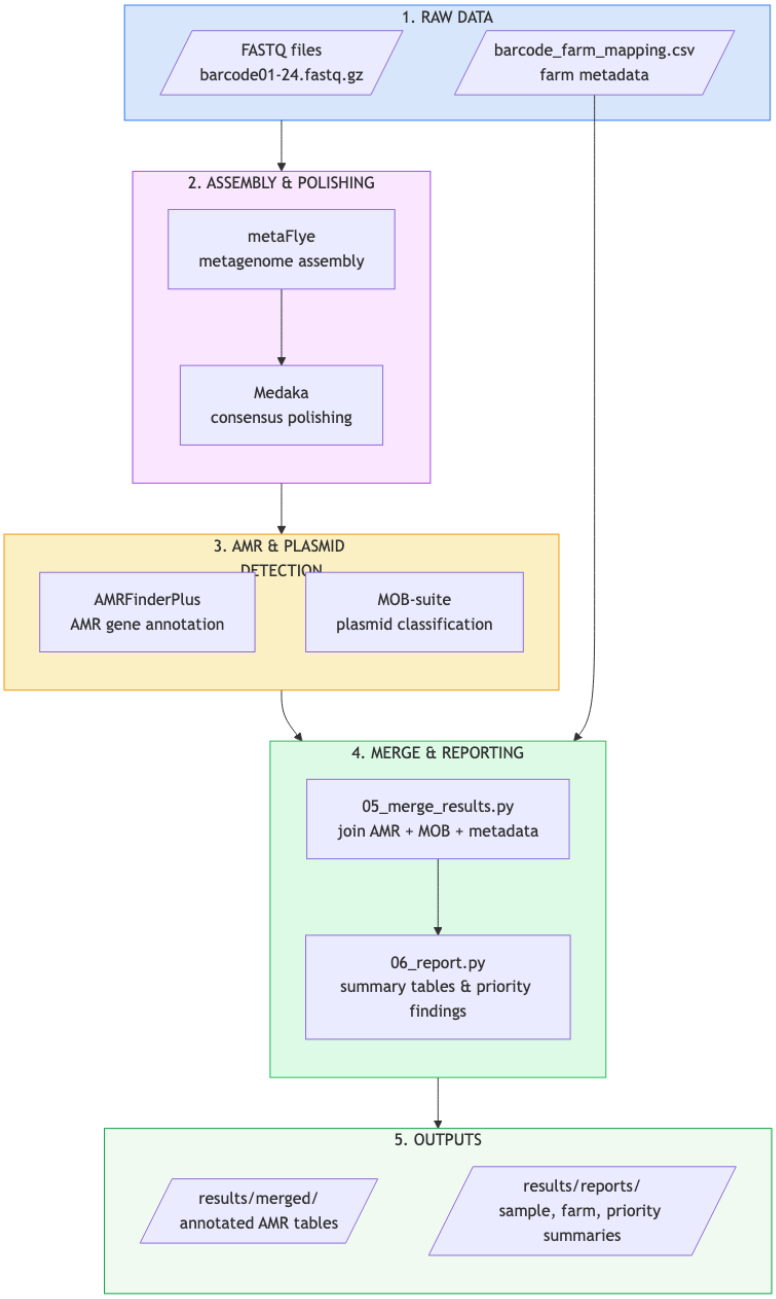
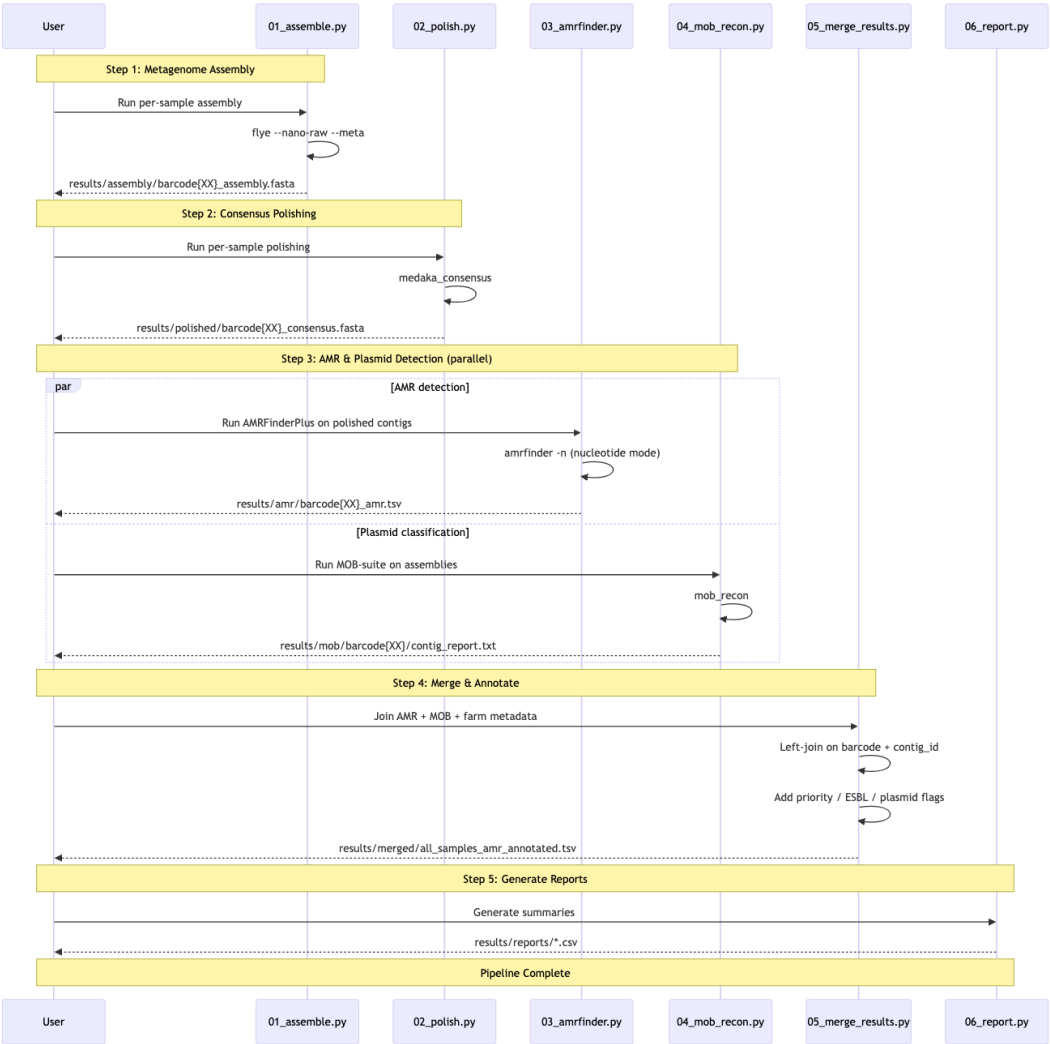
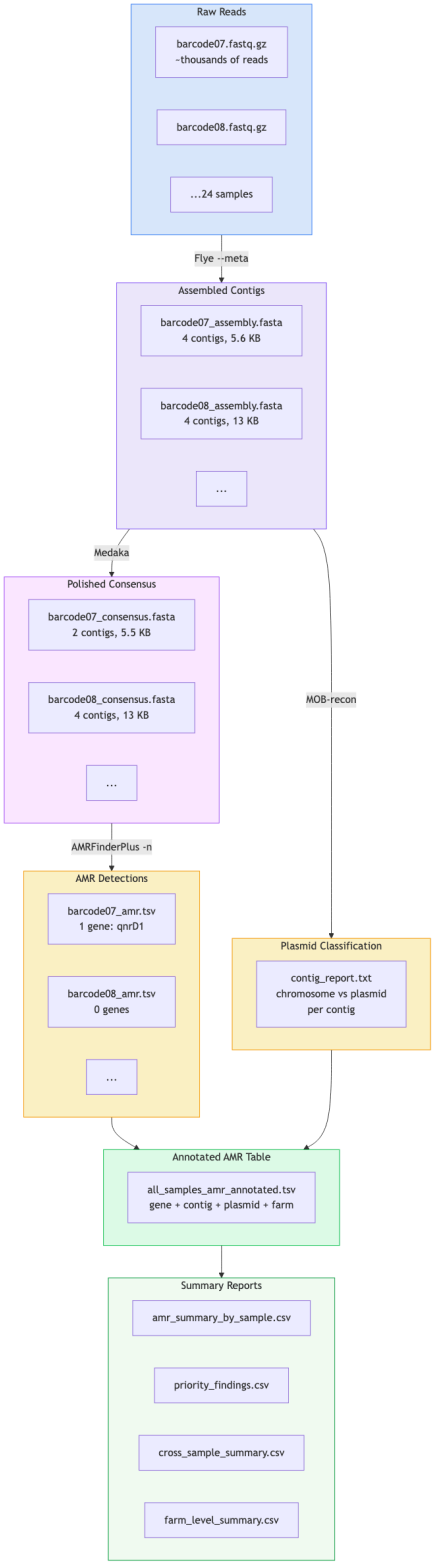
Pipeline Overview


1. RAW DATA
FASTQ files
barcode01-24.fastq.gz
barcode_farm_mapping.csv
farm metadata
2. ASSEMBLY & POLISHING
metaFlye
metagenome assembly
Medaka
consensus polishing
3. AMR & PLASMID DETECTION
AMRFinderPlus
AMR gene annotation
MOB-suite
plasmid classification
4. MERGE & REPORTING
05_merge_results.py
join AMR + MOB + metadata
06_report.py
summary tables & priority findings
5. OUTPUTS
results/merged/
annotated AMR tables
results/reports/
sample, farm, priority summaries

# Data Flow Sequence

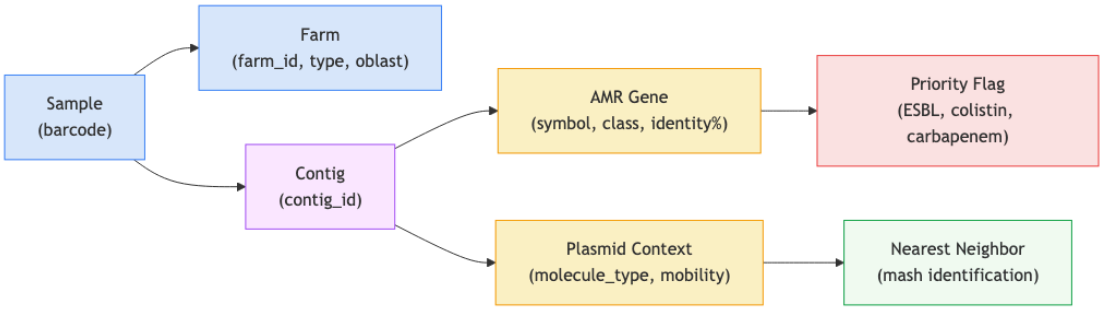


# Data Transformation Diagram



## Evidence Model

Each AMR finding is traceable through the full chain:




*Evidence Model*

## Tool Chain Summary

| Step | Tool | Input | Output | Key Parameters |
|---|---|---|---|---|
| Assembly | metaFlye | Raw FASTQ | Contigs FASTA | `--nano-raw --meta` |
| Polishing | Medaka | FASTQ + Assembly | Consensus FASTA | `-m r1041_e82_400bps_sup_v5.2.0` |
| AMR Detection | AMRFinder Plus | Polished contigs | AMR gene TSV | `-n` (nucleotide mode) |
| Plasmid Typing | MOB-suite | Assembly contigs | Contig report | `mob_recon` |
| Merge | Python/pandas | AMR + MOB + metadata | Annotated table | Left joins on barcode + contig |
| Report | Python/pandas | Annotated table | Summary CSVs | Priority filtering, aggregation |

## Comparison with Main Pipeline

| Aspect | Main Pipeline | PHAGENT_AMR |
|---|---|---|
| AMR source | ResFinder (read-level) | AMRFinderPlus (contig-level) |
| Evidence type | Individual reads | Assembled contigs |
| Plasmid context | None | MOB-suite classification |
| Confidence | Coverage % + Identity % | Assembly + gene coordinates |
| Scope | All 24 barcodes | Currently 4 (scalable to 24) |
| Best for | Broad screening | High-confidence, evidence-backed findings |

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